

Detecting Purpose-Code Gaming in ISO 20022 Payments: A Data-Quality Benchmark and Adversarial Evaluation

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Abstract—From 14 November 2026, cross-border payment messages must carry town and country in designated structured elements, and harmonised data requirements push banks toward registered purpose codes for every transaction. Structured data is expected to improve fraud, sanctions and anti-money-laundering (AML) screening. It also creates a control surface that a payment originator can manipulate. Screening engines examine a defined subset of ISO 20022 elements and score risk partly from the declared purpose code, so an originator who relocates content into an unscreened element, or who declares a lower-scrutiny purpose code, can reduce the chance that a payment is alerted without making the message invalid. Market-practice guidance already records that purpose codes are frequently placed in the wrong element by accident; we treat the deliberate case. We contribute an evasion-primitive taxonomy covering misfielding, code substitution and message-structure substitution over named pain.001 and pacs.008 elements; a data-quality validation framework whose constraints yield detector features, with an open synthetic benchmark generating schema-valid messages at a controllable background rate of *accidental* misfielding alongside labelled adversarial manipulations; and *screening-evasion lift* (SEL), the multiplicative increase in the probability that an illicit payment passes screening un-alerted. The central question the benchmark is built to answer is whether deliberate manipulation is separable from ordinary data-quality error at a realistic noise floor. Measured over 200 000-message corpora at three seeds, it is separable but only weakly, and the separation decays slowly rather than collapsing: the best label-free detector reaches AUC 0.79 when 2% of benign messages carry an accidental defect and 0.74 when 40% do, and at a 1% alert budget it removes almost none of the adversary’s advantage, reducing screening-evasion lift from 1.45–1.78 to 1.44–1.75.

Index Terms—ISO 20022, payments, data quality, sanctions screening, anti-money laundering, adversarial evaluation, benchmark

I. INTRODUCTION

A payment message is a compliance artefact before it is a payment instruction. Sanctions screening reads party names and addresses out of it; AML transaction monitoring reads amounts, counterparties and, increasingly, the declared purpose of the payment. The migration to ISO 20022 changes what those controls can see. Swift removes fully unstructured addresses from cross-border messages on 14 November 2026,

after which town and country must appear in designated elements for all agents and parties [1]. The CPMI harmonised data requirements go further, asking for registered external codes to identify payment type and for structured identification of every party, with an end-2027 implementation horizon [2]. Federal Reserve Financial Services describes the resulting benefit for fraud mitigation in direct terms: standardised category and purpose codes let an institution notice when a payment diverges from a counterparty’s historical pattern [3].

The same change has a second consequence that has attracted much less attention. Screening is selective by design. The Swift guiding principles for screening ISO 20022, endorsed by the Wolfsberg Group, exist precisely to define which elements should and should not be screened, so that banks can screen on a risk basis rather than uniformly [4], [5]. Selectivity is operationally necessary — screening every element of a pacs.008 would be unaffordable in false positives — but it partitions the message into elements that are read by a control and elements that are not. An originator who controls message construction can move content across that partition. Nothing in the schema prevents it. The message remains valid, the payment still settles, and the content a screening engine would have matched on now sits somewhere the engine does not look.

Purpose codes have the same property from the risk-scoring side. If a scoring model treats TRAD as a trade-finance signal warranting extra scrutiny and GDDS as routine, then the choice between them is a choice about expected scrutiny, made by the party with the strongest interest in the outcome. The Payments Market Practice Group already reports that purpose codes are “often misinterpreted by the industry or populated in the incorrect element of a payment instruction,” with knock-on effects on straight-through processing [6]. That is a statement about error. The adversarial version — the same manipulations performed on purpose, to lower alert probability — is discussed in vendor and regulator prose but, as far as our search found, has no peer-reviewed threat model, benchmark or detector.

The gap matters because the two cases are confounded. Accidental misfielding is common, and any detector that flags

it will spend most of its alert budget on data-quality noise. The question worth asking is therefore not whether misfielding can be found — schema validation finds much of it — but whether *deliberate, screening-motivated* manipulation is separable from ordinary error at a realistic noise rate.

Contributions.

- An evasion-primitive taxonomy for ISO 20022 credit transfers, with three families — misfielding, code substitution, message-structure substitution — instantiated on named pain.001 and pacs.008 elements (Section III, Fig. 1). The taxonomy is usable independently of the rest of the paper as a control-design checklist.
- A validation framework of five constraint classes over structured fields, and an open synthetic generator that emits schema-valid pain.001 and pacs.008 instances with a tunable rate of accidental defects and labelled adversarial manipulations (Section IV).
- *Screening-evasion lift*, a metric that separates “the detector caught it” from “the screening engine would have caught it anyway,” and a leave-one-primitive-out protocol that guards against a detector learning the generator rather than the attack (Sections IV and V).

Scope. We do not propose new anomaly-detection mathematics; the detectors are standard and are used as instruments. We use no real payment data: every number in Section VI is measured on the synthetic benchmark and inherits its assumptions.

II. BACKGROUND AND RELATED WORK

A. Fields in scope

We restrict attention to the customer credit transfer pair: pain.001 (initiation, corporate to bank) and pacs.008 (interbank). The elements that matter are `Purp/Cd` and `CtgyPurp/Cd`, both drawn from ISO external code lists; the postal address group `PstlAdr`, with structured children including `Ctry` and `TwnNm` beside the free-text `AdrLine`; the ultimate-party elements `UltmtDbtr` and `UltmtCdtr`; and remittance information, structured or unstructured (`RmtInf/Ustrd`). National authorities publish recommended purpose-code lists for domestic use, narrowing the practical vocabulary in a corridor [7].

B. Screening and its selectivity

The Swift guiding principles [4] and the Wolfsberg payment transparency standards [5] together define what good looks like: screen the elements that carry party identity and jurisdiction, populate them fully, and do not push identifying content into free text. None of this literature models an adversary. It assumes the originator is trying to comply.

C. AML benchmarks

The synthetic-data line of AML research operates at the transaction-graph level. AMLworld provides agent-based synthetic transaction networks with laundering ground truth and is the reference benchmark for graph methods [8]. SAML-D constructs a synthetic transaction monitoring dataset with

typology labels [9]. SISTS adds strategic interaction, letting simulated launderers adapt against a learning detector [10]. These are the closest works to ours in spirit, and the difference is the unit of analysis: they model *who pays whom, how much and when*, with transactions represented as tabular or graph records. None of them represents ISO 20022 message structure, so none can express an attack whose entire mechanism is which element a string is placed in. Transaction-purpose classification has been studied as an enrichment problem — inferring a category from merchant text [11] — but inference of a missing label is a different problem from detecting a label chosen adversarially.

D. Adversarial framing and data quality

The optimisation we use is standard. Adversarial classification as a game between a classifier and a cost-bounded evader dates to Dalvi et al. [13]; the Stackelberg formulation where the defender commits first is due to Brückner and Scheffer [14]; evasion at test time against a deployed model is characterised by Biggio et al. [15]. We reuse these formulations rather than extend them. On the benchmark-design side, Camacho and Rodríguez-Gómez show that for network anomaly detection the choice among dataset variants moves performance more than the choice of learning algorithm [12]. That finding is the methodological reason this paper is organised around a data-quality framework and a controlled noise floor rather than around a model.

E. Positioning

To our knowledge no peer-reviewed work models deliberate manipulation of ISO 20022 structured elements as an evasion strategy, and no public benchmark supplies labelled instances of it. We state this as the outcome of a search over the venues and sources cited here, not as a claim of exhaustive priority.

III. THREAT MODEL AND EVASION TAXONOMY

A. Actors and capabilities

The adversary is the party that constructs the message content: a corporate originator, a payment service provider aggregating client payments, or an insider with access to initiation. This is a weaker assumption than compromising a bank. Constructing a pain.001 with chosen field values is the normal, authorised use of the initiation channel.

The adversary knows the public artefacts: the external code lists, the harmonised data requirements [2], and the guidance on which element classes are screened [4]. A *black-box* adversary knows only that partition and receives a binary accept/hold signal per payment; a *grey-box* adversary also knows the relative risk weights of purpose codes in the corridor, estimable from settlement latency or requests for information. Neither controls the screening engine, the sanctions lists or any bank-side system, and neither sees the alert threshold.

B. Objective

Let x be a message and $\mathcal{M}$ a set of modifications built from the primitives below. Each $m \in \mathcal{M}$ has a cost $c(m)$ combining operational effort and the risk that the manipulation is itself noticed. Let $\pi(x)$ be the probability the payment is alerted by the incumbent screening and monitoring stack, and let $V(x)$ be the predicate “ x is schema-valid, passes the debtor bank’s validation, and still achieves the economic objective.” The adversary solves

$$m^* = \arg \min_{m \in \mathcal{M}} \pi(m(x)) + \lambda c(m) \quad \text{s.t.} \quad V(m(x)), \quad (1)$$

under a budget k on the number of primitives applied. The defender commits to a detector D and threshold τ first, so (1) is the follower’s best response in a Stackelberg game [14]. The constraint V is what makes this problem specific to payments: unlike pixel-space evasion, every modification must survive schema validation, code-list validation, and the receiving bank’s straight-through-processing checks.

C. The primitives

Fig. 1 gives the taxonomy. Three families are distinguished by *what* they change.

Misfielding (P1) changes where content sits without changing what the payment means. Party-name relocation (P1a) moves an ultimate-party name into `RmtInf/Ustrd`, outside the name-screening set. Address destructuring (P1b) writes the real locality into `AdrLine` and puts a benign town and country in the structured children: after November 2026 those children must be present, but nothing requires them to be the most informative values available. Truncation (P1c) abbreviates a name in a screened element below the fuzzy-match threshold, and ultimate-party omission (P1d) drops `UltmtDbtr` where the initiating party is itself a PSP, leaving only the aggregator visible.

Code substitution (P2) keeps the structure and changes the declared semantics. Purpose-code substitution (P2a) selects a code with lower expected scrutiny in the corridor; category-purpose substitution (P2b) does the same for the processing path; code omission (P2c) leaves `Purp` absent where it is optional, forcing the scorer back onto a weaker prior; and code–narrative decoupling (P2d) declares one purpose in the code while the remittance text describes another, which is invisible to any control that reads only the code.

Structure substitution (P3) changes the container. Message-type substitution (P3a) chooses a variant whose screened element set differs; in- message structuring (P3b) splits a value across a batch inside one `pain.001` so no single instructed amount crosses a threshold; party-chain insertion (P3c) adds an intermediary so the party of interest appears only where the next agent does not screen.

P1 and P3 attack *coverage*: the control never sees the content. P2 attacks *scoring*: the control sees everything and assigns the wrong prior. The distinction matters for defence, because coverage attacks leave structural evidence in the message and scoring attacks leave only distributional evidence across messages.

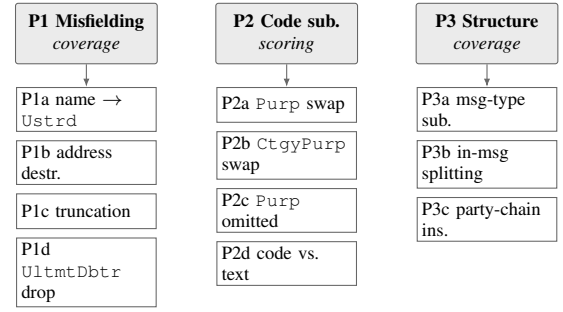


Fig. 1. Evasion primitives for ISO 20022 credit transfers. Families are labelled by the control property they attack: *coverage* means the screening engine never reads the content; *scoring* means it reads the content but is given a misleading prior. Conceptual diagram.

IV. BENCHMARK, VALIDATION FRAMEWORK AND DETECTORS

A. Validation constraints

The framework states five constraint classes over a message x . They are ordered from cheap and local to expensive and longitudinal, and each yields features for the detectors.

C1 Code-list and schema validity. `Purp/Cd` is a member of `ExternalPurpose1Code`; `Ctry` is a valid ISO 3166 alpha-2 code; the instance validates against the message schema.

C2 Structural completeness. The post-2026 minimum (`TwnNm`, `Ctry`) is populated for every agent and party [1], [2]; `AdrLine` does not carry tokens that pattern-match to a structured child, such as a postcode or a country name.

C3 Cross-field consistency. The purpose code is plausible given the creditor’s industry classification and the corridor; `CtgryPurp` agrees with the batch structure; `UltmtDbtr` is present when the initiating party is a PSP.

C4 Code–narrative agreement. The remittance text is consistent with the declared purpose code. We implement this as a similarity between the text and a code description, thresholded per code.

C5 Longitudinal consistency. The purpose-code distribution for a (debtor, creditor, corridor) triple is consistent with its own history. This operationalises the pattern-divergence signal the Federal Reserve article describes [3].

C1 and C2 are hard constraints and produce violations. C3–C5 are soft and produce continuous features. A key property of the design is that C1–C2 violations are dominated by accidental error, while C3–C5 is where the deliberate and accidental cases could plausibly separate.

B. Generator

The generator emits `pain.001.001.09` and `pacs.008.001.08` instances. Schema validity is verified, not asserted: 20 000 messages drawn from a measured corpus, covering all eleven primitives and every accidental-defect mode, validate without error against the ISO 20022 XSDs for both message definitions. It is seeded and deterministic: the same configuration and seed reproduce the same corpus byte-for-byte.

Population. Debtor and creditor entities carry an industry classification, a home country and a corridor membership, and each pair is assigned a purpose-code profile conditioned on industry and corridor.

Benign stream. Messages sample a purpose code from the pair’s profile and populate address and party elements completely. Accidental defects are then injected at rate η , with a defect mix taken from the failure modes that market-practice guidance reports: purpose content written into remittance information, partially destructured addresses, and absent optional codes [6]. η is a benchmark parameter, not a measurement; the protocol sweeps it.

Illicit stream. A fraction ρ of payments involve a watchlist-adjacent party or a high-scrutiny corridor. Without evasion these are the payments the screening abstraction is meant to alert on.

Adversarial stream. For each illicit payment an evader with budget k and knowledge $\kappa \in \{\text{black, grey}\}$ solves (1) against the abstraction π below, by enumeration over admissible primitive subsets. Ground truth records the primitive set applied, its source and target elements, and π before and after.

Screening abstraction. π is a published, parameterised model, not a deployed engine: a name-matching component over the screened element set F_s with a fuzzy-match threshold, plus an additive risk score over purpose code, corridor, amount band and party completeness, alerted at threshold τ calibrated to a stated alert budget. Making π explicit is what makes the evasion metric reproducible, and is also its main limitation (Section VII).

Because SEL is a ratio of pass-through probabilities it is acutely sensitive to the baseline π_0 , so we solve for the parameters rather than choose them. Given a target π_0 — the mean alert probability over *un-evaded* illicit traffic — we bisect the name-match probability p_{name} and, where the score leg puts the target out of reach, the threshold τ , holding $\pi_{\text{floor}} = 0.10$. Targets 0.30, 0.45 and 0.60 solve to $(p_{\text{name}}, \tau) = (0.004, 0.761)$, $(0.004, 0.623)$ and $(0.125, 0.550)$. We report at $\pi_0 = 0.45$ and never quote SEL without the sweep.

C. Screening-evasion lift

Detection accuracy alone does not say whether an attack was worth mounting. Let π_0 be the alert probability for an illicit payment with no manipulation and π_m the alert probability after m . Define

$$\text{SEL}(m) = \frac{1 - \pi_m}{1 - \pi_0}, \quad (2)$$

the multiplicative increase in the probability that an illicit payment passes un-alerted. $\text{SEL} = 1$ means the manipulation bought the adversary nothing. With a detector D in place, the payment must evade both controls, so we also report the residual lift

$$\text{SEL}_{\text{res}}(m, D) = \frac{(1 - \pi_m)(1 - \delta_m)}{(1 - \pi_0)(1 - \delta_0)}, \quad (3)$$

where δ is the detection rate of D at the operating threshold. The gap between (2) and (3) is the part of the attack the

detector actually removes, which is the quantity a control owner needs and is not recoverable from a ROC curve. Both are reported with bootstrap intervals over the corpus.

D. Detectors

Five configurations, from the weakest realistic control upward. D_0 applies C1–C2 only, which is what schema and code-list validation give for free. D_1 adds hand-set thresholds on C3–C4, a rule-based validator of the kind a bank could deploy without a model. D_2 is a supervised gradient-boosted classifier over raw field values — one-hot purpose code, corridor, amount band, completeness flags — trained on gaming labels; it is the naive baseline that assumes labels exist, which in production they largely do not. D_3 is an isolation forest [16] over the C3–C5 constraint features, trained unsupervised on the benign stream only. D_4 combines D_1 and D_3 by equal-weight rank averaging, not by summing z -scores: D_1 is a small-support sum of violations and D_3 a continuous isolation score, and standardising lets the heavier tail dominate (0.775 against 0.768 at $n = 40\,000$). The comparison of interest is D_2 against D_3 : whether constraint features computed without labels match a supervised model that needs them.

V. EVALUATION PROTOCOL

The protocol fixes what will be measured before anything is run.

Corpora. A benign corpus and an adversarial corpus generated from the same population and seeds, so that the only difference is the manipulation. Every result below uses $n = 200\,000$ messages per run across five corridors at seeds 7, 8 and 9, with $\rho = 0.05$ and $\eta = 0.15$ unless swept. The ablations and the η sweep use $n = 100\,000$, the calibration sweep $n = 50\,000$.

Evader parameters. Cost weight $\lambda = 0.18$, rationality temperature $\text{TEMP} = 0.10$; neither is incidental. A perfectly rational evader ($\text{TEMP} = 0$) concentrates on the cheapest primitive and starves the per-primitive analysis, as does raising λ . The softmax keeps all eleven populated, at no fewer than 399 instances each.

Primary metrics. ROC-AUC for gaming detection; detection rate at alert budgets of 1% and 0.1%; and SEL, SEL_{res} per primitive.

Ablations. By primitive family; by evader budget $k \in \{1, 2, 3\}$; by knowledge level κ ; and by the accidental-defect rate η , swept across a range bracketing plausible operational values. The η sweep answers the paper’s question, measuring how fast separability degrades as benign noise rises.

Leave-one-primitive-out. Detectors that learn are trained with one primitive family withheld and evaluated on it. Without this, a detector can score well by learning the generator’s own edit signature rather than the property that makes the edit an attack. We treat the held-out result as the headline number and the in-distribution result as an upper bound.

Reproducibility. Configuration, seeds and library versions are recorded per run; figures regenerate from the raw result files by released scripts.

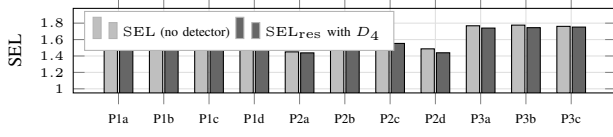


Fig. 2. Screening-evasion lift, seed 7, $\pi_0 = 0.45$; all eleven primitives clear $n \geq 30$ (399–5487), bootstrap half-widths ± 0.021 to ± 0.075 . The gap between the bars is what D_4 removes, and it is small. Mean lift scales with the operating point: 1.40, 1.63, 1.80 at $\pi_0 = 0.30, 0.45, 0.60$.

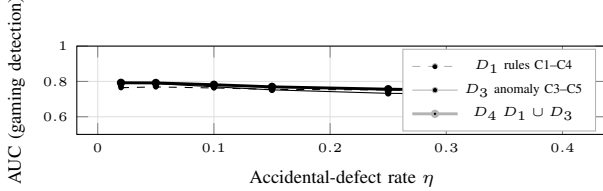


Fig. 3. The experiment the benchmark exists to run: AUC against the rate η at which *benign* messages carry an accidental defect; $n = 100\,000$ per point, seed 7, $\pi_0 = 0.45$. Separability decays without collapsing — D_4 falls 0.792 to 0.742 across a twentyfold rise in η . D_3 learns the benign distribution and is most noise-sensitive (0.793 to 0.724); D_1 is most stable (0.765 to 0.737), overtaking D_3 beyond $\eta \approx 0.15$.

VI. RESULTS

Every number here is measured. Seed spread never exceeds 0.012 AUC.

A. Hypotheses

H1 is not confirmed as stated. It predicted that coverage attacks (P1 and P3) leave structural evidence and would already be caught by D_1 . The families behave nothing alike: against P1 alone D_4 reaches AUC 0.855, against P3 alone 0.519, indistinguishable from chance (Table II). Grouping them was the error. Relocating a name leaves a completeness signature; substituting the message type or inserting an intermediary leaves a structurally ordinary message.

H2 is not confirmed, in either half. It predicted that scoring attacks (P2) leave no single-message evidence, so D_1 would fail on them and D_3 would not. D_1 does not fail: held out on P2 it scores 0.822, its best family. Nor does D_3 overtake it, at 0.796. The premise was wrong — a low-scrutiny purpose code is implausible for the creditor’s industry, which C3 sees, and code–narrative decoupling is visible to C4, so the cross-message route through C5 was never needed.

H3 is confirmed in its second clause only. SEL is not largest for P2a and P1a: at $\pi_0 = 0.45$ the structure family leads (P3b 1.78, P3a 1.77, P3c 1.76), P1a is fourth at 1.74, and P2a is the *smallest* of the eleven at 1.45 — inverted. The second clause holds, SEL_{res} falling furthest for P1a, 1.743 to 1.650.

B. Detectors

D_0 sits at chance: every manipulation leaves a schema-valid, code-list-valid message, so validation alone sees nothing. The premise is measured, not assumed. Among the label-free detectors D_3 does not overtake D_1 in distribution and does so only marginally held out; D_4 leads the three. The comparison

TABLE I
DETECTOR COMPARISON. MEAN OVER SEEDS 7–9, $n = 200\,000$, $\eta = 0.15$, $\pi_0 = 0.45$.

Detector	Lbl.	AUC	@1%	@0.1%	AUC _{LOO}
D_0 schema/code list	no	0.497	0.039	0.039	0.497
D_1 rules C1–C4	no	0.756	0.095	0.042	0.767
D_2 naive classifier	yes	0.997	0.241	0.142	0.841
D_3 anomaly C3–C5	no	0.749	0.017	0.005	0.769
D_4 $D_1 \cup D_3$	no	0.767	0.021	0.007	0.784

Lbl.: needs gaming labels. @ p : TPR at budget p . AUC_{LOO}: leave-one-primitive-family-out over the three families; Section VII states how far it is contaminated.

TABLE II
ABLATION. AUC AND MEAN SEL_{res} FOR D_4 , $n = 100\,000$, SEED 7, $\pi_0 = 0.45$.

Factor	Setting	AUC	SEL_{res}
Primitive family	P1 only	0.855	1.27
	P2 only	0.883	1.37
	P3 only	0.519	1.88
Evader budget	$k = 1$	0.689	1.39
	$k = 2$	0.769	1.61
	$k = 3$	0.833	1.63
Knowledge	black-box	0.674	1.71
	grey-box	0.769	1.61

Section IV set up resolves against the constraint features — D_2 leads on both columns — so labels still buy a great deal. At the budgets a screening operation occupies it is worse: D_4 recovers 2.1% of manipulated messages at a 1% budget and 0.7% at 0.1%, below D_1 ’s 9.5% and 4.2%. Nothing label-free here is deployable.

C. What would have falsified the approach

Neither fired. P2 detection reaches AUC 0.883 against a D_0 baseline of 0.497 at $\eta = 0.15$, so the constraint features do separate deliberate from accidental misfiling; and SEL spans 1.45–1.78 at $\pi_0 = 0.45$ and 1.23–2.19 across the sweep, nowhere near 1, so the abstraction is sensitive to the fields the adversary reaches. The contribution survives; its size, not its existence, disappoints.

VII. DISCUSSION, ETHICS AND THREATS TO VALIDITY

A. Responsible framing

Every primitive in Fig. 1 is derivable from public documents: the external code lists, the harmonised data requirements [2], the guidance stating which element classes are in scope [4], and market-practice guidance recording misfiling as a live problem [6]. We publish no tuned parameters against any deployed engine, and π is a generic model rather than a reproduction of any product. The intended use is control design: a bank can read the taxonomy as a checklist of coverage gaps in its own screened element set. We deliberately withhold corridor- and code-specific weightings that would make the generator an operational recipe.

B. Data and ethics

The corpus is entirely synthetic. No real payment message, customer record or employer data is used, and no proprietary information appears in the generator or its configuration. Entities, accounts and names are drawn from generated distributions, so the release carries no personal data and needs no consent regime. Generator, seeds, configuration and harness are released so others can reproduce or contest the results.

C. Threats to validity

Construct. π is an abstraction of screening, not a screening engine. Every SEL value is therefore relative to a model whose parameters we chose. A different weighting could reorder the primitives. This is the most serious limitation in the paper, and it is not removed by any mitigation available to us without access to a production engine.

Internal. Two measured properties of the generator inflate Table I. The benign stream never batches — every unmanipulated message has $n_{\text{splits}} = 1$ while P3b sets it to 3–7, so $n_{\text{splits}} > 1$ implies manipulation with precision 1.000 and covers 59.9% of manipulated messages. D_2 exploits it and falls from 0.997 to 0.901 when the feature is withheld; real pain.001 files batch routinely, so this is a property of the code, not of payments. It flatters no other detector: D_1 never reads it and D_3 cannot split on a feature constant in training, which is also why P3 alone scores 0.519. Second, leave-one-primitive-family-out is contaminated at $k = 2$, because a message can carry two families and withholding one does not withhold its partner — 99.9% of held-out P1 messages, 77.5% of P3 and 64.1% of P2 still carry a primitive the detector trained on, which is why D_2 barely drops on P1, to 0.991. AUC_{LOO} is an optimistic bound; the clean version needs $k = 1$ and a weaker adversary with it.

External. We do not know the real distribution of purpose codes by corridor and industry, and we do not know the real rate of accidental misfielding. η and the code profiles are assumptions swept over a range, not estimates. Results will not transfer to a bank’s own traffic without recalibration against that traffic.

Conclusion. The protocol has been run, so the detector numbers are measurements, not hypotheses — but measurements on one generator. Three seeds bound the sampling noise and 1000-resample bootstrap intervals bound the SEL estimates; neither touches the two generator properties above, which are why AUC_{LOO} and D_2 are upper bounds. Two of the three hypotheses were not confirmed, which we read as the protocol having been worth running.

D. What we would do next

In the order the results demand: give the benign stream a realistic batch- size distribution and re-run, which removes the n_{splits} artifact and lowers every number in Table I; repeat the ablation at $k = 1$ for an uncontaminated leave-one-out figure; then attack the construct threat, since validating π against an anonymised distribution of alert outcomes from one institution, without moving message data, would make the SEL

numbers calibrated rather than model-relative. Defensive cost stays open, and the false positives a bank would absorb to close each coverage gap decide whether any of this is deployable.

VIII. CONCLUSION

Structured payment data makes screening better and gives the originator a finer instrument for shaping what screening sees. The mandate arriving in November 2026 settles the first half of that trade; the second half has been described in industry guidance as an accident and not yet studied as a choice. We have specified the manipulations available to a message originator, a set of validation constraints over the fields they touch, a synthetic benchmark that generates both accidental and deliberate versions of the same defects, and a metric that separates detector credit from screening credit. Manipulation does still look different from ordinary error at a realistic noise rate. The interesting number is how little that is worth: a detector needing no labels ranks manipulated messages above benign ones about three times in four, and at a 1% alert budget takes back almost none of what the manipulation bought.

Opt plus .5pt Opt

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